INFOGRAPHIC · LANGUAGE MODELS · 2017 → 2026

From *attention* to agents

Nine years separate a Google paper on machine translation from systems that now write code, reason, and operate computers for hours. This is the story in six stages: who entered the game, how the gap between open and closed models closed, which knobs control their behaviour, and what is left of the *stochastic parrot*.

2017
Attention Is All You Need: 8 Google authors propose the Transformer.
VASWANI ET AL. · NEURIPS

175 mil M
GPT-3 parameters, the leap that showed scale alone unlocks abilities.
OPENAI · MAY 2020

~2 months
How long ChatGPT took to reach 100 million users.
UBS ESTIMATE · JAN 2023

8.0% → 1.7%
Lead of the best closed model over the best open one on Chatbot Arena.
JAN 2024 → FEB 2025 · AI INDEX

01 THE TIMELINE

OPEN downloadable weights · CLOSED API / app only · REASONER thinks before answering · MULTIMODAL text + image / audio

STAGE 1 — The foundations

2017 – 2019

An architecture that parallelises on GPU changes what can be trained.

JUN 2017 — Attention Is All You Need
It removes recurrence: attention only. The **Transformer** is born, the base of everything after.
Vaswani et al. · Google

JUN 2018 — GPT-1
Pre-train on unlabelled text, then fine-tune: the *generative pre-trained* pattern.
OpenAI

OCT 2018 — BERT
A bidirectional Transformer; it masters text understanding and reaches Google search in 2019.
Google
OPEN

FEB 2019 — GPT-2
1.5 B parameters. OpenAI withholds the full version over misuse risk: the first public safety debate.
OpenAI · fully released Nov 2019
OPEN

STAGE 2 — The scale bet

2020 – 2022

More data, more parameters, more compute: error falls predictably.

JAN 2020 — Scaling laws
Loss falls as a power law with model size, data, and compute.
Kaplan et al. · OpenAI

MAY 2020 — GPT-3
175 B parameters. It learns tasks from a few examples in the prompt (*few-shot*), without retraining.
OpenAI
CLOSED

2021 — Anthropic is founded
Ex-OpenAI members, with Dario and Daniela Amodei, found a safety-centred lab.
San Francisco

MAR 2021 — «Stochastic Parrots»
It warns of form without meaning, inherited biases, and environmental cost. It names the *stochastic parrot*.
Bender, Gebru, McMillan-Major, Mitchell

MAR 2022 — InstructGPT · Chinchilla
RLHF teaches instruction-following; Chinchilla shows data, not parameters, was missing.
OpenAI · DeepMind

STAGE 3 — The explosion

Nov 2022 – 2023

A chat interface turns a lab technology into a mass product. Everyone joins the race.

30 NOV 2022 — ChatGPT
A chat over RLHF-tuned GPT-3.5. ~100 M users in two months.
OpenAI
CLOSED

FEB 2023 — LLaMA
Meta shares it with researchers; the weights leak in days and the open community takes off.
Meta
OPEN

14 MAR 2023 — GPT-4 and Claude
The same day: GPT-4 takes images as input; Anthropic launches its first Claude.
OpenAI · Anthropic
CLOSED MULTIMODAL

JUL 2023 — Llama 2
Open weights with a commercial-use licence: companies can build on top.
Meta
OPEN

SEP 2023 — Mistral 7B
A French startup, Apache 2.0 licence. A small model that beats 13 B Llama 2.
Mistral AI · Paris
OPEN

DEC 2023 — Gemini 1.0
Natively trained on text, image, audio, and video from the start.
Google DeepMind
CLOSED MULTIMODAL

STAGE 4 — See, hear… and think

2024

Models stop being text-only, and a new lever appears: spending compute while answering.

MAR 2024 — Claude 3
Haiku · Sonnet · Opus family: vision and 200 K tokens of context.
Anthropic
CLOSED MULTIMODAL

MAY 2024 — GPT-4o
Omni: voice, image, and text in one model; real-time voice conversation.
OpenAI
CLOSED MULTIMODAL

JUL 2024 — Llama 3.1 405B
The first open model to compete head-to-head with the closed frontier.
Meta
OPEN

SEP 2024

o1: the first reasoner

It generates a chain of thought before answering. New law: **more inference compute, better answers**.

OpenAI

CLOSED · REASONER

DEC 2024

DeepSeek-V3

A 671 B mixture of experts (37 B active). Final training reported at ~USD 5.6 M of compute.

DeepSeek · Hangzhou

OPEN

STAGE 5

Reasoning opens up

2025

A Chinese lab publishes a cheap open reasoner. The frontier lead is now measured in months.

20 JAN 2025

DeepSeek-R1

An o1-level reasoner with open weights (MIT). On 27 January Nvidia loses ~USD 590 B on the stock market in a day.

DeepSeek

OPEN · REASONER

FEB 2025

Claude 3.7 Sonnet

Hybrid reasoning: the same model answers fast or "thinks" on a user-set budget.

Anthropic

CLOSED · REASONER

APR 2025

Qwen3

An open family (Apache 2.0) with a thinking mode that switches on or off.

Alibaba

OPEN · REASONER

JUL 2025

Kimi K2 · IMO gold

Moonshot opens a 1-trillion-parameter model for agents. OpenAI and DeepMind models reach gold level at the Mathematics Olympiad.

Moonshot · OpenAI · DeepMind

AUG 2025

GPT-5 and gpt-oss

GPT-5 routes between fast and reasoner modes. gpt-oss: OpenAI's first open weights since GPT-2.

OpenAI

REASONER · OPEN (OSS)

NOV 2025

Gemini 3

Google tops the preference rankings at launch; the summit changes hands every few weeks.

Google DeepMind

CLOSED · MULTIMODAL

STAGE 6

From answering to doing

2026

The focus shifts to autonomous work: hour-long tasks, programming, and computer use.

APR 2026

DeepSeek V4

A 1.6-trillion-parameter mixture of experts, 1 M tokens of context, MIT licence. Its self-estimate: 3–6 months behind the closed frontier.

DeepSeek

OPEN · REASONER

JUN – SEP 2026

Claude 5 generation

Fable 5, Sonnet 5, Opus 5, and Opus 5.5: models built for long-running agentic work.

Anthropic

CLOSED · REASONER

SEP 2026

GPT-6 Astra

Staged rollout from 3 September; emphasis on computer use, software engineering, and cybersecurity.

OpenAI

CLOSED · REASONER

02 THE PLAYERS AND THE GAP

Who entered, and when

Founding year of the lab (or of the team behind its model family)

2010 **Google DeepMind**
Gemini · Gemma · merged with Brain in 2023

2015 **OpenAI**
GPT · o-series · gpt-oss

2021 **Anthropic**
Claude

2023 **Meta**
Llama · the open one that opened the door

2023 **Mistral AI**
France · Mistral, Mixtral

2023 **xAI**
Grok

2023 **DeepSeek**
China · V3, R1, V4

2023 **Alibaba · Qwen**
China · Qwen, open since 2023

2023 **Moonshot AI**
China · Kimi

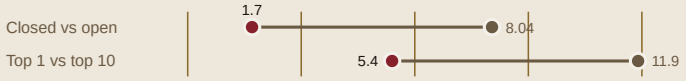
2023 was the entry year: six of these nine teams were born or launched their first LLM in the twelve months after ChatGPT. The frontier stopped belonging to a single lab.

The gap closes

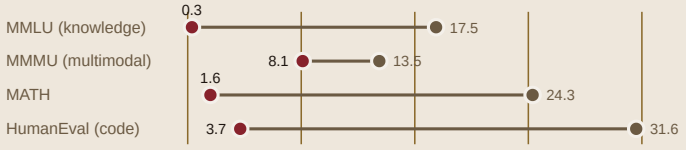
Gap between the leader and its chaser · before and after

before (Jan 2024 / end 2023) · after (Feb 2025 / end 2024)

CHATBOT ARENA · LEADER LEAD (%)



US VS CHINA · BENCHMARK LEAD (PERCENTAGE POINTS)



Source: Stanford HAI, *AI Index 2025*. In 2026 the best open model (DeepSeek V4) sits, by its own estimate, 3–6 months from the closed frontier.

03 THE KNOBS: HOW A MODEL IS CONTROLLED

Temperature

The model does not pick a word: it produces a probability distribution over the next one. Temperature flattens or sharpens it before sampling.

Next word for `The sky is ___`

`T = 0.2`
nearly deterministic

`T = 1`
original distribution

`T = 1.8`
more diversity, more rarity

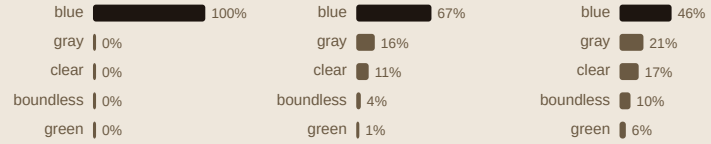
The other knobs

Parameters set on each API call

`top-p` `nucleus sampling` — Only the likeliest tokens are considered until they sum to probability *p* (e.g. 0.9). It cuts the tail of absurd options.

`top-k` — Only the *k* likeliest tokens enter the draw. Coarser than top-p.

`max tokens` — A cap on the answer length, and hence on its cost.



$$p_i = e^{z_i / T} \;/\; \Sigma_j \, e^{z_j / T}$$

z are the *logits*, the raw score the model gives each candidate word. Dividing them by **T** before the *softmax* stretches the differences (T < 1: the favourite almost always wins) or shrinks them (T > 1: rare options gain ground). As T → 0 sampling turns *greedy*: always the likeliest word.

> **Rule of thumb:** low temperature (0–0.3) to extract data, classify, or program; medium (0.7–1) to chat; high (>1) to brainstorm, with a risk of incoherence. The model is the same: what changes is how its options are picked.

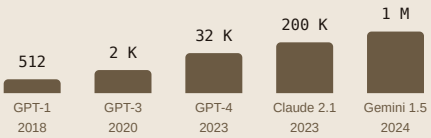
`effort`
`reasoning effort`

In reasoning models, how many "thinking" tokens it spends before answering.

LOW	MEDIUM	HIGH
fast, cheap	balanced	precise, slow

`context`
`visible tokens`

How much text the model "sees" at once. It grew ~2,000×:



> **The 2024–2025 shift:** in reasoners temperature usually arrives fixed or is ignored. The knob that matters becomes **effort**: how much it thinks, not how randomly it picks.

04 THE STOCHASTIC PARROT: DID IT DIE?

THE THESIS · 2021

"A stochastic parrot"

> *A system that stitches sequences of linguistic forms observed in its data, on probabilistic information, without any reference to meaning.*

Bender, Gebru, McMillan-Major, and Mitchell (FAccT 2021). Beyond the conceptual critique, the paper warned of inherited data biases, environmental cost, and the **illusion of understanding** in the reader.

In December 2022, Sam Altman replied: *"i am a stochastic parrot, and so r u"*.

BLOWS AGAINST THE PARROT

Evidence against the thesis

Othello-GPT `Li et al. · 2023`
Trained only on move sequences, the model forms an internal board representation that can be read and edited.

It plans before speaking `Anthropic · 2025`
Interpretability shows Claude picks the word that will rhyme *before* writing the line, and chains intermediate steps on multi-hop questions.

Olympiad mathematics `IMO · 2025`
Proofs written in natural language, at gold-medal level, on unseen problems.

Reasoners `2024 →`
More thinking compute improves answers: behaviour hard to explain as mere repetition.

WHAT IS STILL ALIVE

Evidence for the thesis

Hallucinations
False claims in full fluency. They shrink, but do not vanish.

The reversal curse `Berglund et al. · 2023`
Learning "A is B" does not imply knowing that "B is A".

Unfaithful thinking `Anthropic · 2025`
The visible reasoning chain does not always reflect what truly determined the answer.

Out-of-distribution fragility `Apple · 2025`
The Illusion of Thinking: performance collapses as puzzle complexity rises. A debated result.

VERDICT

Not dead: it got more precise

The strong version, **"there is nothing inside, only surface statistics"**, does not survive the interpretability evidence: there are internal representations and planning. The weak version, **"its reasoning is fragile and its explanation is not always faithful to its computation"**, stays open. Today the useful question is not *whether* the model understands, but *where* and *when* it can be trusted.

SOURCES · Vaswani et al. (2017) *Attention Is All You Need* · Kaplan et al. (2020) *Scaling Laws for Neural Language Models* · Bender et al. (2021) *On the Dangers of Stochastic Parrots* · Hoffmann et al. (2022) *Chinchilla* · Li et al. (2023) *Emergent World Representations* · Berglund et al. (2023) *The Reversal Curse* · DeepSeek-AI (2024–2026) V3, R1, V4 technical reports · Anthropic (2025) *Tracing the thoughts of a large language model* and *Reasoning models don't always say what they think* · Shojaee et al. (2025) *The Illusion of Thinking* · Stanford HAI (2025) *AI Index Report* · 2026 dates checked against announcements and the specialist press (Oct 2026).

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